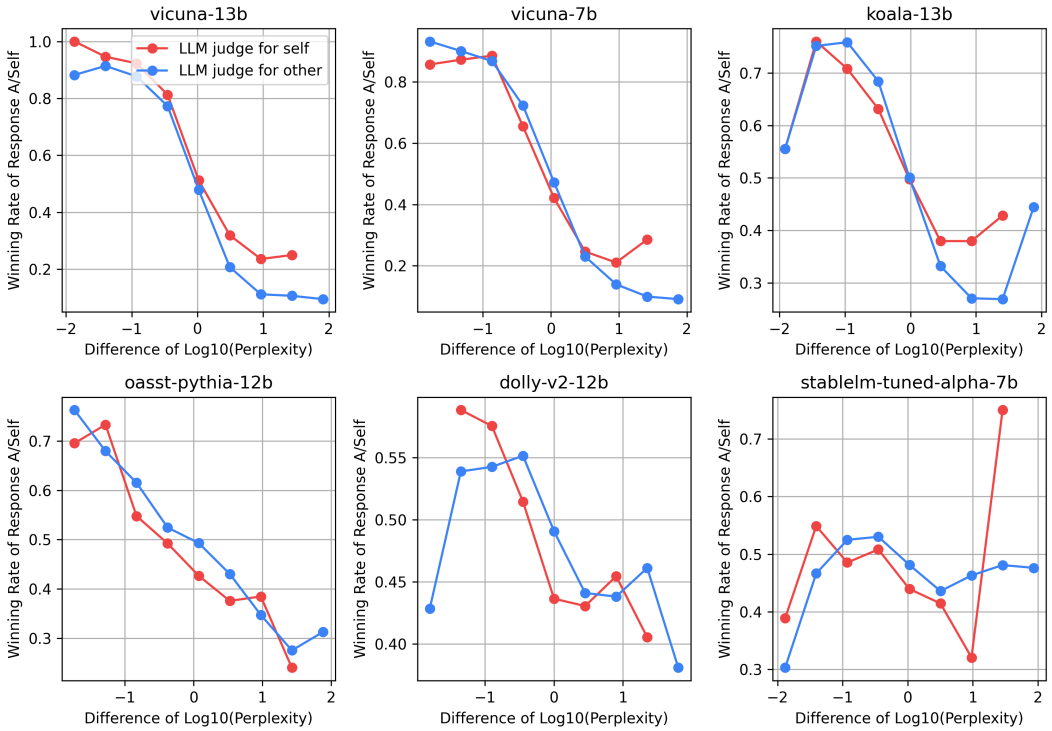

Figure 4: **LLM vs other LLMs conditioned on perplexity.** Winning judgment rates by LLMs on their own texts and texts generated by other models conditioned on perplexity are plotted. Across all models, except for dolly-v2-12b and stablelm-tuned-alpha-7b, no significant difference was observed between the judgment rates for their own texts and those generated by other models. This suggests that LLM evaluators assign higher ratings to texts with lower perplexity, regardless of whether the text was self-generated or produced by other models.

exhibits low perplexity on a sample, decreasing the weight assigned to that model's evaluation for that sample may contribute to bias mitigation. To evaluate the effectiveness of bias reduction strategies, our proposed new metric can be utilized. Therefore, we believe that our research makes a significant contribution to the understanding of self-preference bias and will greatly facilitate the development of future research in this area.

Our experimental results reveal that LLM evaluators tend to assign higher scores to texts with lower perplexity. We further discuss the reasons behind this phenomenon. First, LLMs are trained during the pretraining phase to reduce perplexity on large-scale text corpora. Moreover, when aligning with human preferences, the models are also trained to minimize perplexity on the given dialogue data. Therefore, high-perplexity texts are likely those that the LLM has not frequently encountered during training, suggesting that such texts may be related to domains that the LLM evaluators do not fully comprehend.

This observation may seem contradicted by the fact that GPT-4, which is well-versed across various domains due to a wide range of benchmarks, exhibits a high degree of self-preference bias. However, by investigating specific cases of self-preference bias, as shown in Figure 1a and Table 1, we found that the bias was often not related to clear factual errors but rather to differences in response styles, such as the handling of specialized domains or the description of premises before answering. This suggests that, advanced models like GPT-4, which thoroughly understand and adhere to their predefined policies, may use the degree of alignment with these policies as a deciding factor when evaluating responses of comparable quality.

The proposed metric in Definition 4.1 is based on the fairness definition known as Equal Opportunity. Demographic Parity Calders et al. [2009] is also a prominent definition of fairness. Demographic Parity requires that the predictive distribution of a classifier be consistent across sensitive groups, regardless of the ground truth. In the context of this study, the focus is on whether the evaluations by LLMs align with human preferences, which is why Demographic Parity was not the focal point. While the bias metric based on Demographic Parity cannot demonstrate the unfairness of an evaluation, it is useful for analyzing how highly each LLM evaluator rates its own outputs within the given experimental setup. Similar to the Definition 4.1, the self-preference bias based on Demographic Parity also can be quantified as follows:

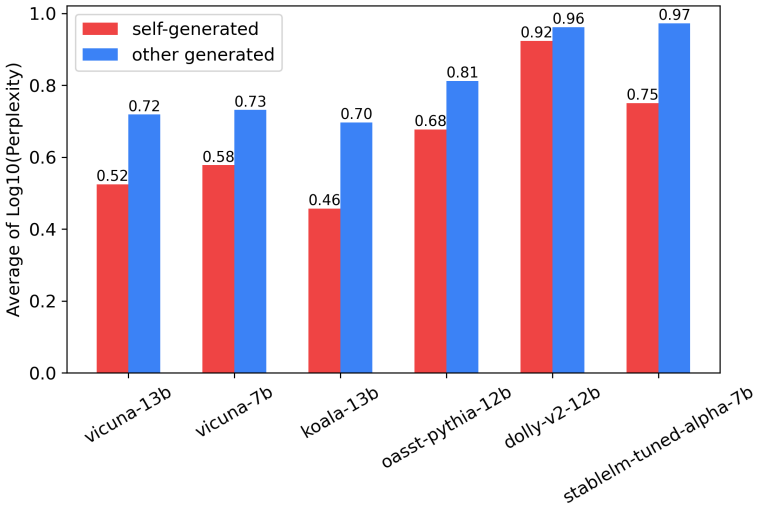

Figure 5: **Average of log-perplexity of responses for each LLM evaluator.** The red bars represent the perplexity for responses generated by the LLM evaluator itself, while the blue bars represent the perplexity for responses generated by other LLMs. Across all models, the average perplexity is lower for responses generated by the evaluators themselves.

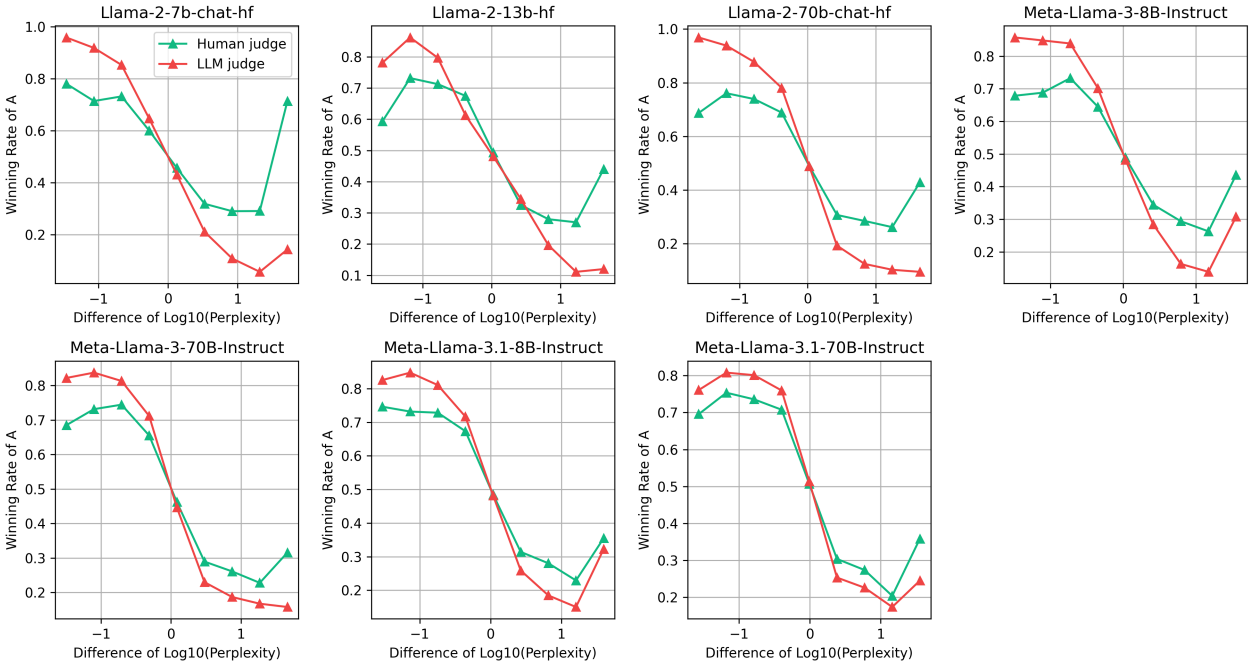

Figure 6: **Llama family vs human conditioned on perplexity.** Winning judgment rates with condition on perplexity are plotted. Llama-2 and later models models exhibited significantly higher winning judgment rate with lower perplexity compared to human evaluators.

$$Bias = P(Y' = 1|S = 1) - P(Y' = 1|S = 0). \tag{4}$$

The scores derived from this metric of eight LLMs are presented in Table 2. The results indicate that GPT-4 exhibited significant bias, followed by Vicuna-13b, which aligns closely with the results obtained using Definition 4.1.

Table 2: **Self-preference bias scores based on Demographic Parity.** GPT-4 assigns the highest scores to its own outputs, followed by Vicuna-13b. However, it is important to note that these scores do not take intrinsic quality into account. Therefore, this analysis reflects how highly each LLM evaluator rates its own outputs within the given experimental setup and should not be interpreted as an indication of unjust evaluation.

LLM	Bias
GPT-4	0.749
GPT-3.5-turbo	0.191
Vicuna-13b	0.382
Vicuna-7b	0.052
Koala-13b	0.175
oasst-pythia-12b	0.006
dolly-v2-12b	-0.069
stablelm-tuned-alpha-7b	-0.032

7 Conclusion

In this study, we propose a metric to quantify the self-preference bias in LLM-as-a-judge and measured the self-preference bias of eight LLMs. Experimental results confirmed that GPT-4, in particular, exhibits a high self-preference bias. This finding suggests a risk that GPT-4 as a judge may inadvertently reinforce its own style and policies.

Furthermore, we hypothesized that the self-preference bias is related to the perplexity of the texts, and showed that, compared to human evaluators, LLM evaluators assigned higher evaluations to texts with lower perplexity, and this tendency was observed regardless of whether the text was generated by themselves or not. This suggests that the essence of the bias lies in perplexity and that the self-preference bias exists because LLMs prefer texts more familiar to them.

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